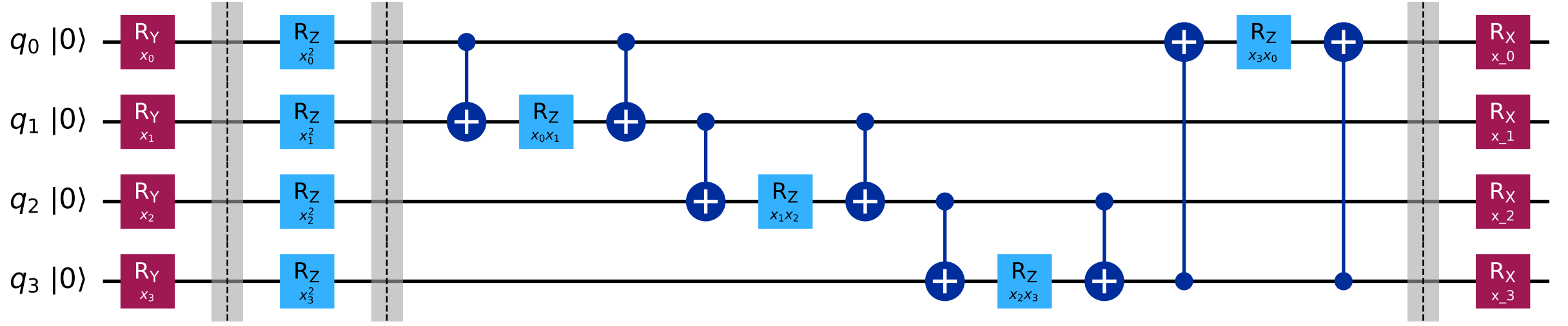
$q_0$ |0⟩
$q_1$ |0⟩
$q_2$ |0⟩
$q_3$ |0⟩
$R_Y$ $x_0$
$R_Y$ $x_1$
$R_Y$ $x_2$
$R_Y$ $x_3$
$R_Z$ $x_0^2$
$R_Z$ $x_1^2$
$R_Z$ $x_2^2$
$R_Z$ $x_3^2$
$R_Z$ $x_0x_1$
$R_Z$ $x_1x_2$
$R_Z$ $x_2x_3$
$R_Z$ $x_3x_0$
$R_X$ x_0
$R_X$ x_1
$R_X$ x_2
$R_X$ x_3